

Personal Information

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Work Experience

Software Engineer > Sim-Ray Engineering Sdn. Bhd. (2024/April~Now)

System Development

- Led the planning and implementation of a web-based in-house ERP system for production, quality, and order tracking.
- Built and supported a Flask-based ERP architecture with PostgreSQL for data management, deployed on a local server for internal factory use.
- Converted real shop-floor workflows into executable system logic and digital processes.
- Managed a small software development team and supported deployment, maintenance, and continuous improvement.

Manufacturing Process Digitalization

- Improved WIP tracking, production status visibility, and cross-department coordination.
- Digitalized quality and compliance-related records to improve traceability and reduce paper- and spreadsheet-based processes.
- Used ERP and process data to identify bottlenecks and support operational improvement.

Engineering Workflow Automation

- Developed an OCR / AI-assisted drawing-processing tool, improving drawing-processing efficiency by approximately 60%.
- Optimized shipment order entry, reporting, and drawing-preparation workflows, improving related efficiency by approximately 50%.
- Reduced downstream production delays caused by documentation and information-preparation bottlenecks.

Master's Research Experience

Embedded Systems & Firmware Development (2021/09~2023/06)

Core Contribution:

- Designed a modular Battery Management System (BMS) architecture for a 20S lithium-titanate battery system, improving system safety, maintainability, and integration readiness.
- Selected the Texas Instruments BQ76942 for voltage, current, and temperature monitoring, and designed protection and cutoff mechanisms for each battery module.

Key Technical Implementation:

- Used STM32F103 as the main controller to communicate with the BQ76942 through SPI, and implemented battery protection logic based on hardware specifications.
- Verified protection mechanisms using charger and load equipment to ensure proper system response under operating and fault conditions.

System Integration & Communication:

- Sent measurement values and protection parameters from STM32 to Orange Pi via UART.
- Integrated multiple BMS data sources for SOC / SOH estimation and remaining energy calculation.
- Transmitted processed battery data through Modbus to the central controller (Raspberry Pi) for higher-level monitoring and control.

Distributed Energy Storage Architecture:

- Built an IEC 61850 communication architecture for battery and inverter integration.
- Modeled Logical Nodes and supported MMS monitoring and GOOSE event communication, enabling system-level data exchange and control coordination.

Education

Exchange: Electrical Engineering - Hanze University of Applied Sciences Groningen -(2023/09~2024/02)

- Optimized Energy Strategy for Sustainable Buildings

Master: Electrical Engineering - National Taipei University of Technology - (2021/09~2023/06)

Thesis: Design of Energy Storage Systems for Distributed Management in Wide Area

- Battery Management System Firmware & Hardware Development
- Associated with Fuel Cell Energy Management System & Energy storage system design

Bachelor: Electrical Engineering - National Taipei University of Technology - (2017/09~2021/06)

- Smart Vehicle Collision Alert System

Language skills

English: Fluent

Chinese: Native

Professional Skills

Programming:

C, Python, JavaScript

Embedded Systems:

STM32, MCU Development, Linux, Embedded System Integration

Interfaces & Protocols:

I2C, SPI, UART, GPIO, CAN, Modbus, MQTT, LoRa, IEC 61850

Debugging & Validation:

Oscilloscope, Logic Analyzer, System Testing, Protection Logic Validation

Software & Data Systems:

PostgreSQL, MySQL, ERP Development, Workflow Automation

Industrial / Energy Systems:

BMS, EMS, Embedded Linux Platforms